

Processing of mutant cystic fibrosis transmembrane conductance regulator is temperature-sensitive.

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Cystic fibrosis transmembrane conductance regulator (CFTR) is a plasma membrane Cl⁻ channel regulated by cyclic AMP-dependent phosphorylation and by intracellular ATP. Mutations in CFTR cause cystic fibrosis partly through loss of cAMP-regulated Cl⁻ permeability from the plasma membrane of affected epithelia. The most common mutation in cystic fibrosis is deletion of phenylalanine at residue 508 (CFTR delta F508) (ref. 10). Studies on the biosynthesis and localization of CFTR delta F508 indicate that the mutant protein is not processed correctly and, as a result, is not delivered to the plasma membrane. These conclusions are consistent with earlier functional studies which failed to detect cAMP-stimulated Cl⁻ channels in cells expressing CFTR delta F508 (refs 16, 17). Chloride channel activity was detected, however, when CFTR delta F508 was expressed in *Xenopus* oocytes, Vero cells and Sf9 insect cells. Because oocytes and Sf9 cells are typically maintained at lower temperatures than mammalian cells, and because processing of nascent proteins can be sensitive to temperature, we tested the effect of temperature on the processing of CFTR delta F508. Here we show that the processing of CFTR delta F508 reverts towards that of wild-type as the incubation temperature is reduced. When the processing defect is corrected, cAMP-regulated Cl⁻ channels appear in the plasma membrane. These results reconcile previous contradictory observations and suggest that the mutant most commonly associated with cystic fibrosis is temperature-sensitive.

Altered biogenesis of deltaF508-CFTR following treatment with doxorubicin.

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Cystic fibrosis (CF) is caused by mutations to the cystic fibrosis transmembrane conductance regulator (CFTR) gene. The most common of these mutations is deletion of a phenylalanine residue at position 508 (Delta F508), which accounts for approximately 70% of all CF alleles. This mutation interferes with the biogenesis and maturation of Delta F508-CFTR to the plasma membrane. However, Delta F508-CFTR can partially function upon proper localization. Thus, pharmacological correction of Delta F508-CFTR maturation holds promise in CF therapy. Our previous studies indicate that a single non-cytotoxic dose of the anthracycline doxorubicin (Dox) significantly increase Delta F508-CFTR-associated chloride secretion in MDCK cells by increasing the expression of this protein at the apical plasma membrane. We report here that Dox alters the biogenesis of Delta F508-CFTR. Treatment with Dox increases the resistance of Delta F508-CFTR to trypsin digestion, possibly by expediting protein folding. Further, treatment with Dox reduces the amount of polyubiquitinated Delta F508-CFTR in cells and prolongs the half-life of this protein. Concomitantly, treatment with Dox decreases the association of Delta F508-CFTR with HSP70 but does not alter the expression of major HSP70 family members. Based on these results, we propose that Dox expedites the folding and maturation of Delta F508-CFTR by acting as a pharmacological chaperone, which consequently promotes the functional expression of this protein in MDCK cells.

A new missense mutation G126D in exon 4 of the cystic fibrosis transmembrane conductance regulator (CFTR) gene.

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Cystic fibrosis (CF) is the most common severe autosomal recessive disorder in the Caucasian population. Beside the major mutation delta F508, which accounts for approximately 68% of all CF chromosomes, more than 350 different point mutations leading to this disease have been detected and communicated to the 'Cystic Fibrosis Genetic Analysis Consortium'. As these mutation are scattered about the whole CFTR gene we used denaturing gradient gel electrophoresis as a rapid method for screening a large number of CF patients for point mutations in the CFTR exons.

Cystic fibrosis gene variability in two southern Brazilian Amerindian populations: analysis of the deltaF508 mutation and the KM19 and XV2C haplotypes.

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The frequencies of the deltaF508 deletion, the most common cystic fibrosis mutation in Europeans and European-derived populations, and the XV2C and KM19 restriction fragment length polymorphisms that are tightly linked to the CFTR locus vary among populations. To determine the distribution of these extragenic markers and of the deltaF508 mutation, we analyzed 326 chromosomes of individuals from two South American Indian populations, the Guarani and the Kaingang. The allele and haplotype frequencies differed greatly between the two populations as well as among Amerindians and normal European Brazilians and European Brazilian cystic fibrosis patients. The absence of the deltaF508 mutation and the B haplotype are in agreement with the hypothesis that the deltaF508 mutation occurred after the divergence of these two populations. This finding is useful for populations containing a large Amerindian component and helps us to understand the origins of the deltaF508 deletion, the most common cystic fibrosis mutation in Europeans and European-derived populations, as well as the different incidences of cystic fibrosis in continental groups.

Heterozygous UCHL1 loss-of-function variants cause a neurodegenerative disorder with spasticity, ataxia, neuropathy, and optic atrophy.

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Biallelic variants in UCHL1 have been associated with a progressive early-onset neurodegenerative disorder, autosomal recessive spastic paraplegia type 79. In this study, we investigated heterozygous UCHL1 variants on the basis of results from cohort-based burden analyses. Gene-burden analyses were performed on exome and genome data of independent cohorts of patients with hereditary ataxia and spastic paraplegia from Germany and the United Kingdom in a total of 3169 patients and 33,141 controls. Clinical data of affected individuals and additional independent families were collected and evaluated. Patients' fibroblasts were used to perform mass spectrometry-based proteomics. UCHL1 was prioritized in both independent cohorts as a candidate gene for an autosomal dominant disorder. We identified a total of 34 cases from 18 unrelated families, carrying 13 heterozygous loss-of-function variants (15 families) and an inframe insertion (3 families). Affected individuals mainly presented with spasticity (24/31), ataxia (28/31), neuropathy (11/21), and optic atrophy (9/17). The mass spectrometry-based proteomics showed approximately 50% reduction of UCHL1 expression in patients' fibroblasts. Our bioinformatic analysis, in-depth clinical and genetic workup, and functional studies established haploinsufficiency of UCHL1 as a novel disease mechanism in spastic ataxia.

Commonly Reported Mosquito-Borne Viruses in the United States: A Primer for Pharmacists.

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Mosquito-borne diseases are a public health concern. Pharmacists are often a patient's first stop for health information and may be asked questions regarding transmission, symptoms, and treatment of mosquito borne viruses (MBVs). The objective of this paper is to review transmission, geographic location, symptoms, diagnosis and treatment of MBVs. We discuss the following viruses with cases in the US in recent years: Dengue, West Nile, Chikungunya, LaCrosse Encephalitis, Eastern Equine Encephalitis Virus, and Zika. Prevention, including vaccines, and the impact of climate change are also discussed.

Historical Hot Spots of Dengue and Zika Viruses to Guide Targeted Vector Control in San Juan, Puerto Rico (2010-2022).

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Dengue viruses (DENV) continue to cause large outbreaks in tropical countries, while chikungunya and Zika (ZIKV) viruses have added complexity to Aedes-borne disease prevention and control efforts. Because these viruses are transmitted by the same vectors in urban areas, it is useful to understand if sequential outbreaks caused by these viruses have commonalities, such as similar seasonal and spatial patterns, that would help anticipate and perhaps prevent future outbreaks. We explored and analyzed the heterogeneity of confirmed cases of DENV (2010-2014 and 2015-2022) and ZIKV (2016-2017) during outbreaks in the San Juan metropolitan area of Puerto Rico to explore their degree of overlap and prioritize areas for Aedes aegypti control. Deidentified, georeferenced case data were aggregated into grid cells (500 × 500 m) within a geographical information system of the study area and analyzed to calculate the degree of overlap between outbreaks. Spatial autocorrelations using local indicators of spatial associations were conducted to identify significant disease case hot spots and correlations between outbreaks. We found that 75% of cases during the three transmission periods were concentrated in 25% of the total number of grid cells covering the study area. We also found significant clustering of cases during each outbreak, enabling identification of consistent disease hot spots. Our results showed 85% spatial overlap between cases of ZIKV in 2015-2017 and DENV in 2010-2014 and 97% overlap between DENV cases in 2010-2014 and 2015-2022. These results reveal urban areas at greater risk of future arbovirus outbreaks that should be prioritized for vector control.

High-resolution mapping of urban Aedes aegypti immature abundance through breeding site detection based on satellite and street view imagery.

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Identification of Aedes aegypti breeding hotspots is essential for the implementation of targeted vector control strategies and thus the prevention of several mosquito-borne diseases worldwide. Training computer vision models on satellite and street view imagery in the municipality of Rio de Janeiro, we analyzed the correlation between the density of common breeding grounds and Aedes aegypti infestation

measured by ovitraps on a monthly basis between 2019 and 2022. Our findings emphasized the significance ($p \leq 0.05$) of micro-habitat proxies generated through object detection, allowing to explain high spatial variance in urban abundance of *Aedes aegypti* immatures. Water tanks, non-mounted car tires, plastic bags, potted plants, and storm drains positively correlated with *Aedes aegypti* egg and larva counts considering a 1000 m mosquito flight range buffer around 2700 ovitrap locations, while dumpsters, small trash bins, and large trash bins exhibited a negative association. This complementary application of satellite and street view imagery opens the pathway for high-resolution interpolation of entomological surveillance data and has the potential to optimize vector control strategies. Consequently it supports the mitigation of emerging infectious diseases transmitted by *Aedes aegypti*, such as dengue, chikungunya, and Zika, which cause thousands of deaths each year.

Exposure to Zika and chikungunya viruses impacts aspects of the vectorial capacity of *Aedes aegypti* and *Culex quinquefasciatus*.

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Zika (ZIKV) and chikungunya (CHIKV) are arboviruses that cause infections in humans and can cause clinical complications, representing a worldwide public health problem. *Aedes aegypti* is the primary vector of these pathogens and *Culex quinquefasciatus* may be a potential ZIKV vector. This study aimed to evaluate fecundity, fertility, survival, longevity, and blood feeding activity in *Ae. aegypti* after exposure to ZIKV and CHIKV and, in *Cx. quinquefasciatus* exposed to ZIKV. Three colonies were evaluated: AeCamp (*Ae. aegypti*-field), RecL (*Ae. aegypti*-laboratory) and CqSLab (*Cx. quinquefasciatus*-laboratory). Seven to 10 days-old females from these colonies were exposed to artificial blood feeding with CHIKV or ZIKV. CHIKV caused reduction in fecundity and fertility in AeCamp and reduction in survival and fertility in RecL. ZIKV impacted survival in RecL, fertility in AeCamp and, fecundity and fertility in CqSLab. Both viruses had no effect on blood feeding activity. These results show that CHIKV produces a higher biological cost in *Ae. aegypti*, compared to ZIKV, and ZIKV differently alters the biological performance in colonies of *Ae. aegypti* and *Cx. quinquefasciatus*. These results provide a better understanding over the processes of virus-vector interaction and can shed light on the complexity of arbovirus transmission.

The rising global economic costs of invasive *Aedes* mosquitoes and *Aedes*-borne diseases.

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Invasive *Aedes aegypti* and *Aedes albopictus* mosquitoes transmit viruses such as dengue, chikungunya and Zika, posing a huge public health burden as well as having a less well understood economic impact. We present a comprehensive, global-scale synthesis of studies reporting these economic costs, spanning 166 countries and territories over 45 years. The minimum cumulative reported cost estimate expressed in 2022 US\$ was 94.7 billion, although this figure reflects considerable underreporting and underestimation. The analysis suggests a 14-fold increase in costs, with an average annual expenditure of US\$ 3.1 billion, and a maximum of US\$ 20.3 billion in 2013. Damage and losses were an order of magnitude higher than investment in management, with only a modest portion allocated to prevention. Effective control measures are urgently needed to safeguard global health and well-being, and to reduce

the economic burden on human societies. This study fills a critical gap by addressing the increasing economic costs of Aedes and Aedes-borne diseases and offers insights to inform evidence-based policy.